Solutions to the Challenges in *Chaos: An Introduction to Dynamical Systems*

Alligood, Sauer, Yorke — Part I: Challenges 1, 2 and 3

Two tools used throughout

Lemma 1 (exact pull-back)**.** *Let g be continuous on a compact interval J and let $T \subseteq g(J)$ be a compact interval. Then J contains a compact interval J' with $g(J') = T$.*

Proof. Write $T = [t_1, t_2]$; if $t_1 = t_2$ any point of $g^{-1}(t_1)$ works. Otherwise pick $a, b \in J$ with $g(a) = t_1$, $g(b) = t_2$; say $a < b$ (the other case is symmetric). Let $u = \max\{x \in [a, b] : g(x) = t_1\}$ and $v = \min\{x \in [u, b] : g(x) = t_2\}$. If some $x \in (u, v)$ had $g(x) \leq t_1$, the intermediate value theorem on $[x, v]$ would give a point with $g = t_1$ beyond u, contradicting the choice of u; if $g(x) \geq t_2$, we would get a point of $[u, x]$ with $g = t_2$, contradicting the choice of v. So $g([u, v]) = [t_1, t_2]$. □

Lemma 2 (covering)**.** *(a) If g is continuous on a compact interval J and $g(J) \supseteq J$, then g has a fixed point in J.*
(b) Let $J_0, \ldots, J_{m-1}$ be compact intervals with $f(J_i) \supseteq J_{i+1}$ for $0 \leq i \leq m-2$ and $f(J_{m-1}) \supseteq J_0$. Then there is $x \in J_0$ with $f^i(x) \in J_i$ for $0 \leq i < m$ and $f^m(x) = x$.

Proof. (a) With $J = [s, t]$ there are $p, q \in J$ with $g(p) = s \leq p$ and $g(q) = t \geq q$; so $g(x) - x$ changes sign on J. (b) By Lemma 1 choose $Q_{m-1} \subseteq J_{m-1}$ with $f(Q_{m-1}) = J_0$; then $Q_{m-2} \subseteq J_{m-2}$ with $f(Q_{m-2}) = Q_{m-1}$ (possible since $f(J_{m-2}) \supseteq J_{m-1} \supseteq Q_{m-1}$); and so on down to $Q_0 \subseteq J_0$ with $f(Q_0) = Q_1$. Then $f^m(Q_0) = J_0 \supseteq Q_0$, so by (a) f^m has a fixed point $x \in Q_0$, and $f^i(x) \in Q_i \subseteq J_i$. □

1 Challenge 1: Period three implies chaos

Set-up and conventions

Let $A < B < C$ with $f(A) = B$, $f(B) = C$, $f(C) = A$, and let f be unimodal as in Figure 1.14: there is a critical point $m \in (A, B)$ with f increasing on $[A, m]$ and decreasing on $[m, C]$ (and $f(m) > C$). Put $L = [A, B]$, $R = [B, C]$, $h = C - A$.

- f is decreasing on R with $f(B) = C$, $f(C) = A$, so f maps R monotonically onto $[A, C]$. Let $x^* \in (B, C)$ be the point with $f(x^*) = B$. Then $RR = [B, x^*]$ (points of R mapped into R) and $RL = [x^*, C]$ (points of R mapped into L). Put $d = \text{length}(RR) = x^* - B$.
- f is increasing on $[A, m]$ from $f(A) = B$ up to $f(m) > C$. Let $a_1 \in (A, m)$ with $f(a_1) = C$. The points of L that are mapped into $[A, C]$ are exactly $[A, a_1]$ (plus the single point B, which is the period-three point mapped to C and which we ignore, as the text does). Thus $LR = [A, a_1]$, and f maps LR monotonically onto R.

For a word $w = S_1 \cdots S_m$ in $\{L, R\}$ with no two consecutive L's let

$$I(w) = \{x : \ f^{\,i-1}(x) \in S_i, \ 1 \le i \le m\}.$$

Because f is monotone on R and on $[A, a_1]$, induction on m gives:

(F1) $I(w)$ is a closed interval and $f^{\,j}$ maps it monotonically onto $I(S_{j+1} \cdots S_m)$; in particular $f^{m-1}(I(w)) = S_m$. (If $f(S_i) \supseteq S_{i+1}$ at every stage, every point of $I(S_{i+1} \cdots S_m)$ has a preimage in the right piece of S_i.)

(F2) $I(wu) \subseteq I(w)$ for every admissible extension u.

(F3) Different words of the same length give intervals with disjoint interiors (they differ at some symbol and L, R meet only at B); they can touch only at endpoints.

Note $I(RR) = [B, x^*]$, $I(RL) = [x^*, C]$, $I(LR) = [A, a_1]$.

A point x has *sensitive dependence* if there is $\delta > 0$ such that for every $\varepsilon > 0$ there are y with $|y - x| < \varepsilon$ and n with $|f^n(x) - f^n(y)| \ge \delta$.

Step 1. Let $J = I(S_1 \cdots S_k R)$ and write $u = S_1 \cdots S_k R$. Then $I(uRL) = I(S_1 \cdots S_k RRL) \subseteq J$ and $I(uLR) = I(S_1 \cdots S_k RLR) \subseteq J$ (both words are admissible because $R \to R$, $R \to L$, $L \to R$ are allowed). By (F1), f^{k+1} maps $I(uRL)$ onto $I(RL) = [x^*, C]$ and $I(uLR)$ onto $I(LR) = [A, a_1]$. Hence for *every* $x \in I(uRL)$ and $y \in I(uLR)$,

$$|f^{k+1}(x) - f^{k+1}(y)| \ \ge \ x^* - a_1 \ > \ x^* - B = d.$$

In particular J contains a pair of points, one in each of the two required subintervals, that are eventually more than d apart.

Step 2. Inside $J = I(u)$ ($u = S_1 \cdots S_k R$) the words $uS_{k+2}S_{k+3}R$ with $S_{k+2}S_{k+3} \in \{RR, RL, LR\}$ are admissible (LL is not), giving three subintervals $I(uRRR), I(uRLR), I(uLRR)$ of J. They have positive length (a power of f maps each onto an interval of positive length) and, by (F3), disjoint interiors. If two of them had length $\ge \frac{1}{2}\,\mathrm{length}(J)$ they would already fill J, leaving no room for the (positive-length) third. Hence at least two of the three have length $< \frac{1}{2}\,\mathrm{length}(J)$.

Step 3. Let $K = I(uSSR)$ be one of the two short intervals of Step 2, $|K| < \frac{1}{2}|J|$, where $uSSR = S_1 \cdots S_k\, R\, S_{k+2}S_{k+3}R$ has $k' + 1 = k + 4$ symbols. Put

$$J_1 = I(uSSR\,LR) = I(S_1 \cdots S_k R S_{k+2}S_{k+3}RLR) \ \subseteq K.$$

Let $x \in J_1$ be arbitrary and pick any $y \in I(uSSR\,RL) \subseteq K$ (nonempty by (F1)). Then $|x - y| \le |K| < \frac{1}{2}\,\mathrm{length}(J)$, and applying Step 1 to the word $uSSR$ (which ends in R) shows $|f^{k+4}(x) - f^{k+4}(y)| > d$. So every point of J_1 has a neighbour within $\mathrm{length}(J)/2$ that eventually separates from it by more than d.

Step 4. Start with the single interval $I(R) = R$, of length $C - B < h$. Given an interval $I(u)$ with u ending in R, Steps 2–3 give two disjoint-interior subintervals $I(uSSR\,LR)$, one for each of two short choices of SS, each ending in R, with five more symbols and length less than $\frac{1}{2}|I(u)|$. After n rounds we have 2^n intervals with disjoint interiors, each named by $5n + 1$ symbols and of length $< 2^{-n}h$.

Each contains a point with sensitive dependence. Take a nested chain $J^{(0)} \supset J^{(1)} \supset \cdots$ of such intervals (one from each round) and let x be the point in all of them (the lengths tend to 0). At round n the point x lies in $J_1 = I(uSSR\,LR) \subseteq K = I(uSSR)$ for the appropriate u, K, so by

Step 3 there is $y_n \in K$ with $|x - y_n| < 2^{-n}h$ and some iterate with $|f^{N_n}(x) - f^{N_n}(y_n)| > d$. As $n \to \infty$, $y_n \to x$, so x has sensitive dependence with $\delta = d$. Since $2^n \to \infty$, there are infinitely many sensitive points.

Step 5. Fix for each interval $I(u)$ an ordering of its two chosen children (say left, right) and call them child 0 and child 1. To a binary sequence $a = .a_1a_2a_3\cdots$ associate the nested chain $I_0 \supset I_1 \supset \cdots$ with I_n the child a_n of I_{n-1}, and the point $x(a) = \bigcap_n I_n$. By Step 4, $x(a)$ is sensitive. Different sequences first differ at some level n, where the two intervals have disjoint interiors; so $x(a) \neq x(a')$ unless $x(a)$ is a common endpoint of two intervals of the same level, i.e. a point with some iterate in $\{A, B, C\}$. A point lies in at most two intervals of a given level, so at most two sequences a give the same point. Hence $a \mapsto x(a)$ is at most 2-to-1 and its image is uncountable. (This matches the usual looseness of the binary expansions $.0111\ldots = .1000\ldots$.) So the set of sensitive points is uncountable.

Step 6. The three points A, B, C can be ordered on the line in $3! = 6$ ways, so there are 5 other cases. List the orbit points in increasing order as $a < b < c$. Either (I) $f(a) = b,\ f(b) = c,\ f(c) = a$ (this is the case treated, with letters A, B, C) or (II) $f(a) = c,\ f(c) = b,\ f(b) = a$.

- Orders $A < B < C$, $B < C < A$, $C < A < B$ are all of type (I): they are the same picture with the letters renamed (start the cycle at the smallest point).
- The other three orders are of type (II). Conjugating by the reflection $\varphi(x) = -x$, i.e. $\tilde{f}(x) = -f(-x)$, turns a type (II) map into a type (I) map ($\tilde{f}(-c) = -b,\ \tilde{f}(-b) = -a,\ \tilde{f}(-a) = -c$), and φ is an isometry, so sensitive dependence is preserved.

So essentially one argument suffices. Within type (I) there is one more variant: the unimodal map may have its maximum in R instead of L (the gap in the domain then moves from LR to R). Then f is increasing on all of L (mapping it onto R), and on R it increases up to the maximum and then decreases to A. The set RR is the part of the decreasing branch mapped into R, RL the part mapped into L, and RL still lies to the right of RR. Every step above goes through with the same $d = \text{length}(RR)$; in Step 1 one even gets a larger separation since $LR = L$ lies to the left of B.

Step 7. Let f be any continuous map with a period-three orbit; by Step 6 take $a < b < c$ with $f(a) = b, f(b) = c, f(c) = a$, and set $L = [a, b]$, $R = [b, c]$. We replace the (possibly many-lapped) pieces by single "one-piece" subintervals mapped *exactly* onto L and R:

- $\Lambda \subseteq L$ with $f(\Lambda) = R$: take $u = \max\{x \in L : f(x) = b\}$ and $v = \min\{x \in [u, b] : f(x) = c\}$ and $\Lambda = [u, v]$ (Lemma 1).
- $P \subseteq R$ with $f(P) = R$ and $Q \subseteq R$ with $f(Q) = L$, with P to the left of Q and disjoint interiors: let $r = \min\{x \geq b : f(x) = b\}$ ($> b$ since $f(b) = c \neq b$), $w = \max\{x \in [b, r] : f(x) = c\}$, $P = [w, r]$; let $r' = \max\{x \in [b, c] : f(x) = b\} \geq r$, $z = \min\{x \geq r' : f(x) = a\}$, $Q = [r', z]$.

Then $f(\Lambda) = R$, $f(P) = R$, $f(Q) = L$, and $\Lambda \subseteq [a, b]$, $Q \subseteq [r', c]$ with $r' > b$. Set $\delta = r' - b > 0$; every point of Λ is at least δ from every point of Q.

Call a closed interval J an *R-interval of depth n* if $f^n(J) = R$. Lemma 1 (applied to f^n) lets us pull back inside J any prescribed closed target inside R. Use it as follows.

- (Step 1) Pick $P_2 \subseteq P$ with $f(P_2) = Q$ and $Q_2 \subseteq Q$ with $f(Q_2) = \Lambda$. Pull back P_2 and Q_2 into J to get $J' \subseteq J$ with $f^n(J') = P_2$ and $J'' \subseteq J$ with $f^n(J'') = Q_2$. Then $f^{n+1}(J') = Q$, $f^{n+1}(J'') = \Lambda$, $f^{n+2}(J'') = R$; so J'' is an R-interval of depth $n + 2$, and any $x \in J'$, $y \in J''$ satisfy $|f^{n+1}(x) - f^{n+1}(y)| \geq \delta$.

- (Step 2) Choose three closed targets $T_1, T_2, T_3 \subseteq R$ with pairwise disjoint interiors, each satisfying $f^3(T_i) = R$ (one inside P using the branch $P \to P \to P \to R$, one inside Q using $Q \to \Lambda \to P \to R$, and one inside P using $P \to Q \to \Lambda \to R$, taken so that the targets of the second step are disjoint). Pulling them back into J gives three R-intervals of depth $n+3$ with disjoint interiors; at least two have length $< |J|/2$ exactly as before.
- (Steps 3–5) The argument is then word-for-word the one above, with d replaced by δ and "$I(w)$" replaced by the pulled-back intervals (nested by construction).

So any continuous map with a period-three orbit has uncountably many points with sensitive dependence (hence also those of Step 5), as claimed. ■

2 Challenge 2: Periodic orbits of linear maps on a torus

Throughout $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has integer entries and $\det A \neq 0$, $S(v) = Av \pmod 1$ on $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$, and Fibonacci numbers are indexed as in the book, $F_0 = F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$ (also $F_{-1} = 0$).

Step 1. A is linear and $A\mathbb{Z}^2 \subseteq \mathbb{Z}^2$. If $v \equiv v'$ modulo $\mathbb{Z}^2$, then $Av - Av' = A(v - v') \in \mathbb{Z}^2$, so $Av \equiv Av'$: the map $v \mapsto Av \pmod 1$ is well defined on $\mathbb{R}^2/\mathbb{Z}^2$. Concretely, $A(0, y)^T = (by, dy)$ and $A(1, y)^T = (a + by, c + dy)$ differ by the integer vector (a, c), so the two (identified) points $(0, y), (1, y)$ have the same image mod 1; likewise $(x, 0), (x, 1)$ differ by (b, d) in the image. The composition $\mathbb{R}^2 \xrightarrow{A} \mathbb{R}^2 \to \mathbb{T}^2$ is continuous and constant on the classes of the quotient, hence S is continuous on $\mathbb{T}^2$.

Step 2. (a) $A(x + n_1, y + n_2)^T = A(x, y)^T + A(n_1, n_2)^T$ and $A(n_1, n_2)^T \in \mathbb{Z}^2$, so the two sides agree mod 1. (b) $S^2(v) = A(Av - m) \pmod 1$ for some $m \in \mathbb{Z}^2$ and $A(Av - m) = A^2v - Am \equiv A^2v$ by (a). (c) If $S^n(v) \equiv A^n v$, then $S^{n+1}(v) = A(A^n v - m) \equiv A^{n+1}v$ by (a), where $m \in \mathbb{Z}^2$. So only the last step needs "mod 1".

Step 3. Suppose 1 is not an eigenvalue, i.e. $\det(A - I) \neq 0$. If $S(v) = v$ then $Av = v + n$ with $n \in \mathbb{Z}^2$, so $v = (A - I)^{-1}n$. The inverse $(A - I)^{-1} = \operatorname{adj}(A - I)/\det(A - I)$ has rational entries, so v is rational.

Step 4. Assume no eigenvalue has modulus one. Then A^n has no eigenvalue 1 for every $n \geq 1$.

($\Rightarrow$) If v is eventually periodic, then $S^m(v)$ is periodic for some m, say $S^p(S^m v) = S^m v$. By Step 2(c), $S^m v$ is a fixed point of the torus map given by the integer matrix A^p, which has no eigenvalue 1, so by Step 3 $S^m v$ is rational. Also $S^m v = A^m v - n$ with $n \in \mathbb{Z}^2$, so $v = A^{-m}(S^m v + n)$ is rational (A^{-m} has rational entries).

($\Leftarrow$) If $v = (p_1/q, p_2/q)$, then every iterate has coordinates in $\frac{1}{q}\mathbb{Z}$ (since A is an integer matrix), so all iterates lie in the finite set $\{(i/q, j/q) : 0 \leq i, j < q\}$. By the pigeonhole principle $S^i(v) = S^j(v)$ for some $i < j$, so v is eventually periodic.

Step 5. Fix $v_0 \in \mathbb{T}^2$. Then S is a continuous group homomorphism of the compact group $\mathbb{T}^2$, and it is onto because $A : \mathbb{R}^2 \to \mathbb{R}^2$ is onto. Each fibre $S^{-1}(v_0)$ is therefore a coset of $\ker S$, and all fibres have the same number N of points.

To compute N: the solutions of $S(v) = v_0$ in $[0, 1)^2$ correspond to the integer vectors n with $A^{-1}(v_0 + n) \in [0, 1)^2$, i.e. with $v_0 + n \in \Pi := A([0, 1)^2)$, the half-open parallelogram spanned by (a, c) and (b, d). So $N(v_0) = \#\big((v_0 + \mathbb{Z}^2) \cap \Pi\big) = \sum_{n \in \mathbb{Z}^2} \mathbf{1}_\Pi(v_0 + n)$. Integrate over $v_0 \in [0, 1)^2$:

$$N = \int_{[0,1)^2} \sum_n \mathbf{1}_\Pi(v_0 + n)\, dv_0 = \int_{\mathbb{R}^2} \mathbf{1}_\Pi = \operatorname{area}(\Pi) = |\det A|.$$

So $S(v) = v_0$ has exactly $|\det A|$ solutions for every v_0.

Step 6. $S(v) = v$ means $(A - I)v \equiv 0 \pmod 1$, i.e. $S_{A-I}(v) = 0$ for the torus map of the integer matrix $A - I$. If $\det(A - I) \neq 0$, Step 5 (with $v_0 = 0$) gives exactly $|\det(A - I)|$ solutions, and

$$\det(A - I) = (a - 1)(d - 1) - bc = \det A - \operatorname{tr} A + 1.$$

So the number of fixed points is $|\det A - \operatorname{tr} A + 1|$ whenever this is nonzero.

Step 7. Let $B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Induction gives $B^n = \begin{pmatrix} F_n & F_{n-1} \\ F_{n-1} & F_{n-2} \end{pmatrix}$ (use $B^{n+1} = B\,B^n$ and the Fibonacci recursion), and $A = B^2$. A direct check is also easy: $A^1 = \begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix}$ and

$$A \begin{pmatrix} F_{2n} & F_{2n-1} \\ F_{2n-1} & F_{2n-2} \end{pmatrix} = \begin{pmatrix} 2F_{2n} + F_{2n-1} & 2F_{2n-1} + F_{2n-2} \\ F_{2n} + F_{2n-1} & F_{2n-1} + F_{2n-2} \end{pmatrix} = \begin{pmatrix} F_{2n+2} & F_{2n+1} \\ F_{2n+1} & F_{2n} \end{pmatrix}.$$

Hence $A^n = \begin{pmatrix} F_{2n} & F_{2n-1} \\ F_{2n-1} & F_{2n-2} \end{pmatrix}$.

Step 8. Here $\det A = 1$, $\operatorname{tr} A = 3$, so $|\det A - \operatorname{tr} A + 1| = 1$: the origin is the only fixed point. For S^2 we use $A^2 = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}$, $\det(A^2 - I) = 4 \cdot 1 - 9 = -5$, so S^2 has 5 fixed points, i.e. 4 points of period two (two orbits). Solving $(A^2 - I)v = n \in \mathbb{Z}^2$ gives $v = \frac{1}{5}\begin{pmatrix} -1 & 3 \\ 3 & -4 \end{pmatrix} n \equiv \frac{1}{5}\begin{pmatrix} 4 & 3 \\ 3 & 1 \end{pmatrix} n$, so the solutions mod 1 are the multiples of $(.8, .6)$:

$$(0, 0),\ (.8, .6),\ (.6, .2),\ (.4, .8),\ (.2, .4).$$

Check: $S(.2, .4) = (.8, .6)$, $S(.8, .6) = (.2, .4)$, $S(.4, .8) = (.6, .2)$, $S(.6, .2) = (.4, .8)$. So the period-two orbits are $\{(.2, .4), (.8, .6)\}$ and $\{(.4, .8), (.6, .2)\}$.

Step 9. S^n is the torus map of A^n, $\det A^n = 1$, $\operatorname{tr} A^n = F_{2n} + F_{2n-2}$, so by Step 6 the number of fixed points of S^n is

$$N_n = |2 - F_{2n} - F_{2n-2}| = \left|(F_{2n} - 1)(F_{2n-2} - 1) - F_{2n-1}^2\right|,$$

using $F_{2n}F_{2n-2} - F_{2n-1}^2 = \det A^n = 1$. For $n = 1, \ldots, 5$ this gives $1, 5, 16, 45, 121$.

Step 10. Let $D_n = F_nF_{n-2} - F_{n-1}^2$. Then $D_2 = 2 \cdot 1 - 1 = 1$ and

$$D_{n+1} = F_{n+1}F_{n-1} - F_n^2 = (F_n + F_{n-1})F_{n-1} - F_n^2 = F_{n-1}^2 - F_nF_{n-2} = -D_n,$$

so $D_n = (-1)^n$. For the second identity ($F_{2n} = F_n^2 + F_{n-1}^2$) take the $(1, 1)$ entry of $B^{2n} = B^nB^n$, where B^n is symmetric. Now

$$F_{2n}+F_{2n-2} = \big(F_n^2+F_{n-1}^2\big)+\big(F_{n-1}^2+F_{n-2}^2\big) = F_n^2+2\big(F_nF_{n-2}-(-1)^n\big)+F_{n-2}^2 = (F_n+F_{n-2})^2-2(-1)^n,$$

and the sum is $\geq 3 > 2$, so

$$N_n = F_{2n} + F_{2n-2} - 2 = (F_n + F_{n-2})^2 - 2 - 2(-1)^n.$$

(Remark: $F_n + F_{n-2}$ is the Lucas number L_n, so $N_n = L_{2n} - 2$.)

Step 11. The number P_n of points of least period n is $P_n = \sum_{d|n} \mu(n/d)N_d$ (Möbius inversion) and the number of orbits is P_n/n. The values $N_n = L_{2n} - 2$ are

$$N_1, \ldots, N_{10} = 1,\ 5,\ 16,\ 45,\ 121,\ 320,\ 841,\ 2205,\ 5776,\ 15125,$$

and the periodic table is

period n	1	2	3	4	5	6	7	8	9	10
fixed points of S^n	1	5	16	45	121	320	841	2205	5776	15125
points of least period n	1	4	15	40	120	300	840	2160	5760	15000
orbits of period n	1	2	5	10	24	50	120	270	640	1500

(For example $P_6 = N_6 - N_3 - N_2 + N_1 = 320 - 16 - 5 + 1 = 300$.) This agrees with Figure 2.31: $45 - 5 = 40$ period-four points (ten orbits), $121 - 1 = 120$ period-five points (24 orbits), $16 - 1 = 15$ period-three points.

Every period occurs. For $n \geq 3$, the points of period $< n$ dividing n number at most $\sum_{d \leq n/2} N_d < \sum_{d \leq n/2} L_{2d} = L_{2\lfloor n/2 \rfloor + 1} - 1 \leq L_{n+1}$, while $N_n = L_{2n} - 2 \geq L_{n+2} - 2 = L_{n+1} + L_n - 2 > L_{n+1}$ for $n \geq 3$. Hence $P_n > 0$. Also $P_1 = 1$, $P_2 = 4$.

Step 12. Use $A^n = B^{2n}$ and $B^{2n} - I = B^n(B^n - B^{-n})$. Since $B^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$ is an integer matrix, $B^{\pm n}$ preserves $\mathbb{Z}^2$, so $(B^{2n} - I)v \in \mathbb{Z}^2 \iff (B^n - B^{-n})v \in \mathbb{Z}^2$. Using $\det B = -1$,

$$B^{-n} = (-1)^n \begin{pmatrix} F_{n-2} & -F_{n-1} \\ -F_{n-1} & F_n \end{pmatrix}.$$

n odd: $B^n - B^{-n} = (F_n + F_{n-2})\, I$. So the fixed points of S^n are exactly v with $bv \in \mathbb{Z}^2$, $b = F_n + F_{n-2}$: all points $(i/b, j/b)$, $b^2 = N_n$ of them (consistent with Step 10).

n even: using $F_n - F_{n-2} = F_{n-1}$, $B^n - B^{-n} = F_{n-1}C$ with $C = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$, $C^2 = 5I$. So $F_{n-1}Cv = w \in \mathbb{Z}^2$ gives $v = C^{-1}w/F_{n-1} = Cw/(5F_{n-1})$: all fixed points of S^n have the form $(i/b, j/b)$ with $b = 5F_{n-1}$ (and there are $5F_{n-1}^2 = N_n$ of them).

Since points of period n are fixed points of S^n, they are all of the form a/b with $b = F_n + F_{n-2}$ for n odd and $b = 5F_{n-1}$ for n even, as claimed. For $n = 2$: $b = 5F_1 = 5$ (matches $(.2, .4), \ldots$); $n = 3$: $b = F_3 + F_1 = 4$; $n = 4$: $b = 5F_3 = 15$ (matches Figure 2.31(a)); $n = 5$: $b = 11$ (matches the 11×11 grid); $n = 6$: $b = 5F_5 = 40$. ■